

1. Strong, A.B., Plastic Materials and Processing, Prentice Hall, New Delhi (1999)
2. Middleman, S., Fundamentals of Polymer Processing, Houghton Mifflin Company, UK(1997)

Reference Books

1. Chawala, K.K., Composite materials, Springer-Verlag, New York (1987)
2. Tadmor, Z. and Gogos, C.G., Principles of Polymer Processing, John Wiley, US (2006).

Evaluation Scheme

S. No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessionals (Research Assignments/Projects/Tutorials/Quizes)	25

UME730 : Computer Aided Manufacturing

L	T	P	Cr
2	0	2	3.0

Course Objective: To introduce the students to the standard terminologies, conventions, processes, design and operational characteristics of key hardware components, programming techniques and applications of modern multi-axis computer numerical control (CNC) machining centers. To expose the students to automatic computer assisted CNC tool path programming and virtual simulation of toolpath data for CNC milling and turning centres using modern professional software.

Syllabus

Introduction: fundamental concepts in computer numeric control (CNC): types, definition and designation of control axes, special constructional and design characteristics of CNC machine tools, standard tooling for CNC turning and milling centres, types and functions of

CNC systems, advantages of CNC technology compared to conventional manufacturing, types and functions DNC (direct numeric control), advantages of combined CNC/DNC systems.

CNC part programming: Work holding and tool setting procedures for CNC turning and milling centres, tool zero pre-setting, tool length and radius compensation, manual part programming including use of standard canned cycles for CNC turning and milling centres, introduction to automatic part programming using standard CAM software.

Computer numerical control elements: introduction to sensors, drives and feedback devices used in CNC systems, control loop circuit elements in point to point (PTP) and contouring system, interpolation schemes for linear and circular interpolations, types and functions adaptive control systems.

Introduction to advanced CNC machining systems: Advantages of 3-1/2-1/2 axis, 4-axis and 5-axis CNC machining systems, types of 5-axis machining centres, advanced CNC controllers and their special features for multi-axis machining, fundamentals of automatic toolpath planning for multi-axis CNC machining systems,

Automated manufacturing support systems: introduction to group technology, introduction to automated quality control in manufacturing, automated material transfer, handling, storage and identification systems: AGVs, ASRS, carousel, and RFID technologies.

Laboratory Work

Exercises in tool pre-setting, work piece referencing and manual part programming for machining of simple parts on CNC turning and milling centres, use of CAM software for simulation of turning and milling toolpath data for simple parts, automatic cutter location data generation from CAD models and post-processing for machining on CNC machines using standard CAD-CAM software, and use of CMM for automatic quality control.

Students in a group of 5/6 will carry out project on generation of CNC part program for machining of a part shape in milling or turning centre using standard canned cycles. Each student group will submit a report on the procedure followed for executing the given assignment along with the part machined on specified CNC machining centre.

Course Learning Objectives (CLO)

The students will be able to:

1. Create a plan for machining of a given part on a multi-axis CNC machining center including selection of machining parameters, cutting tools, process sequence and controller settings for tool presets.
2. Create and validate a CNC part program data using manual data input (MDI) / commercial CAM package for machining of component using a CNC machining centre.